

RATIONAL HOMOTOPY THEORY IN GEOMETRY – ASSIGNMENT

- (1) Describe bigraded model for a cohomologically 1–connected graded algebra. Let

$$H = \Lambda(x_1, x_2, x_3, x_4)/I$$

be a graded algebra over $\mathbb{Q}$, where

$$|x_1| = |x_2| = |x_3| = 3, \quad |x_4| = 5,$$

and the ideal I is generated by

$$I = \langle x_1x_2, x_1x_3x_4, x_2x_3x_4 \rangle.$$

Let

$$(\Lambda Z, d) \xrightarrow{\simeq} (H, 0)$$

be the bigraded (Halperin–Stasheff) model of H . Compute Z_0, Z_1, Z_2 .

- (2) Define Sullivan representative of a continuous map $f : (Y, *) \rightarrow (X, *)$, where X, Y simply connected. Define the linear part $Q(f)$ of this representative. Let $(\Lambda V, d)$ be the minimal Sullivan model of a simply connected space X . Define a pairing

$$(-; -) : V \times \pi_*(X) \longrightarrow \mathbb{Q}$$

by the rule

$$(v; \alpha)e = \begin{cases} Q(\alpha)(v), & v \in V^{|\alpha|}, \\ 0, & \deg v \neq \deg \alpha, \end{cases}$$

where e denotes the generator of $H^*(S^{|\alpha|}; \mathbb{Q})$, and $Q(\alpha) : (\Lambda V, d) \rightarrow (\Lambda(e), 0)$ is the morphism induced by $\alpha \in \pi_*(X)$.

Prove that this map is bilinear. Show that for all $v \in V$ and $\beta \in \pi_*(Y)$, the pairing satisfies

$$(Q(f)v; \beta) = (v; \pi_*(f)(\beta)).$$

- (3) Show that suspension of a well based topological space is always formal.
- (4) Define the Whitehead product. Let $a \in \pi_{2n}(S^{2n})$ be the generator. Consider the Whitehead product $[a, a]_W \in \pi_{4n-1}(S^{2n})$.
- (a) Show that $[a, a]_W$ is non-trivial.
- (b) Prove that $[a, a]_W$ is not a torsion element in $\pi_{4n-1}(S^{2n})$.

Read chapter 13 of Felix, Halperin, Thomas at least these parts.